Self Learnign Assignment Day 3
C++ Programming

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Write a C++ program for a Vehicle Registration System using inheritance, runtime polymorphism,
templates, and file handling.

1) Abstract Base Class: Vehicle

Properties: RegistrationNumber, ModelName, Manufacturer, Year.

Pure virtual function: displayInfo().

Validation: Year should be between 2000 and the current year.

2) Derived Classes
Car

Adds property: FuelType

Bike

Adds property: EngineCapacity

3) Other Requirements

Use runtime polymorphism (Vehicle* or smart pointers) to manage both Car and Bike objects.

Use a template function to validate the input year.

Use a static data member in Vehicle to count how many vehicle records are created.

Use an inline function to format and return the vehicle display string.

Store vehicle objects into a binary file, then read them back and display the records.

Implement exception handling for invalid inputs (e.g., empty registration number, invalid year).

Sample Input
Enter number of vehicles: 2

Enter type of vehicle (C for Car, B for Bike): C
Enter Registration Number: MH20AB1234
Enter Model Name: Swift
Enter Manufacturer: Maruti
Enter Year: 2018
Enter Fuel Type: Petrol

Enter type of vehicle (C for Car, B for Bike): B
Enter Registration Number: MH20XY5678
Enter Model Name: Pulsar
Enter Manufacturer: Bajaj
Enter Year: 2020
Enter Engine Capacity: 150cc
Expected Output (after reading from file)
RegNo: MH20AB1234, Model: Swift, Manufacturer: Maruti, Year: 2018, FuelType: Petrol
RegNo: MH20XY5678, Model: Pulsar, Manufacturer: Bajaj, Year: 2020, EngineCapacity: 150cc
----
Total Vehicles stored: 2

Program:

```
#include <iostream>
#include <fstream>
using namespace std;

// Template function to check year
template <class T>
bool checkYear(T y)
{
   if (y >= 2000 && y <= 2025)
       return true;
   else
       return false;
}

// Base Class
class Vehicle
{
protected:
   string regNo, model, manufacturer;
   int year;

public:
```

```
    static int count;

    Vehicle(string r, string m, string manu, int y)
    {
        if (r == "")
            throw "Registration number cannot be empty";

        if (!checkYear(y))
            throw "Invalid year";

        regNo = r;
        model = m;
        manufacturer = manu;
        year = y;

        count++;
    }

    // inline function
    inline string basicInfo()
    {
        return "RegNo: " + regNo +
               ", Model: " + model +
               ", Manufacturer: " + manufacturer +
               ", Year: " + to_string(year);
    }

    virtual void displayInfo() = 0;
};

int Vehicle::count = 0;

// Car Class
class Car : public Vehicle
{
    string fuelType;

public:
    Car(string r, string m, string manu, int y, string fuel)
        : Vehicle(r, m, manu, y)
    {
        fuelType = fuel;
    }

    void displayInfo()
```

```
    {
        cout << basicInfo() << ", FuelType: " << fuelType << endl;
    }
};

// Bike Class
class Bike : public Vehicle
{
    string engineCap;

public:
    Bike(string r, string m, string manu, int y, string cap)
        : Vehicle(r, m, manu, y)
    {
        engineCap = cap;
    }

    void displayInfo()
    {
        cout << basicInfo() << ", EngineCapacity: " << engineCap << endl;
    }
};

int main()
{
    int n;
    cout << "Enter number of vehicles: ";
    cin >> n;

    Vehicle *v[10];

    ofstream file("vehicles.txt");

    try
    {
        for (int i = 0; i < n; i++)
        {
            char type;
            string reg, model, manu;
            int year;

            cout << "\nEnter type (C for Car, B for Bike): ";
            cin >> type;

            cout << "Enter Registration Number: ";
```

```
            cin >> reg;

            cout << "Enter Model Name: ";
            cin >> model;

            cout << "Enter Manufacturer: ";
            cin >> manu;

            cout << "Enter Year: ";
            cin >> year;

            if (type == 'C')
            {
                string fuel;
                cout << "Enter Fuel Type: ";
                cin >> fuel;

                v[i] = new Car(reg, model, manu, year, fuel);
            }
            else
            {
                string cap;
                cout << "Enter Engine Capacity: ";
                cin >> cap;

                v[i] = new Bike(reg, model, manu, year, cap);
            }

            v[i]->displayInfo();

            file << v[i]->basicInfo() << endl;
        }

        file.close();

        cout << "Records saved in file!! " << endl ;
        cout << "Total Vehicles stored: " << Vehicle::count << endl;
    }
    catch (const char *msg)
    {
        cout << "Error: " << msg << endl;
    }

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day3_AbhinayKumar(260240120005) % cd "/Users/abhinyaay/
60240120005)/" && g++ vecReg.cpp -o vecReg && "/Users/abhinyaay/Desktop/cpp /SLA_Day3_AbhinayKuma
Enter number of vehicles: 3

Enter type (C for Car, B for Bike): C
Enter Registration Number: UP84AF5136
Enter Model Name: dzire
Enter Manufacturer: maruti
Enter Year: 2020
Enter Fuel Type: petrol
RegNo: UP84AF5136, Model: dzire, Manufacturer: maruti, Year: 2020, FuelType: petrol

Enter type (C for Car, B for Bike): b
Enter Registration Number: up84d1052
Enter Model Name: passionplus
Enter Manufacturer: herohonda
Enter Year: 2005
Enter Engine Capacity: 95
RegNo: up84d1052, Model: passionplus, Manufacturer: herohonda, Year: 2005, EngineCapacity: 95

Enter type (C for Car, B for Bike): b
Enter Registration Number: up80fw1737
Enter Model Name: passionpro
Enter Manufacturer: herhonda
Enter Year: 2021
Enter Engine Capacity: 110
RegNo: up80fw1737, Model: passionpro, Manufacturer: herhonda, Year: 2021, EngineCapacity: 110
Records saved in file!!
Total Vehicles stored: 3
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day3_AbhinayKumar(260240120005) %
```

Output File:
vehicles.txt

RegNo: UP84AF5136, Model: dzire, Manufacturer: maruti, Year: 2020
RegNo: up84d1052, Model: passionplus, Manufacturer: herohonda, Year: 2005
RegNo: up80fw1737, Model: passionpro, Manufacturer: herhonda, Year: 2021